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Constraint-Based Problem Solving: Hospital Doctor Shift Scheduling

Course: Artificial Intelligence (CSA17) | **Assessment Tool 2** | **CO1**

1. Problem Understanding

A hospital must assign 3 doctors to 3 shifts. This is a Constraint Satisfaction Problem (CSP), defined by variables, domains and constraints.

Variables: D1, D2, D3 (each doctor's shift)

Domain of each variable: {Morning (M), Afternoon (A), Night (N)}, with the shift order M < A < N

Constraints:

No.	Constraint	Type
C1	D1 ≠ Night	Unary
C2	D2's shift is before D3's shift (D2 < D3)	Binary
C3	Only one doctor per shift (D1, D2, D3 all different)	All-different
C4	D3 ≠ Morning	Unary
C5	Every doctor gets exactly one shift	Completeness

C5 is satisfied automatically because each variable takes exactly one value from its domain.

2. Constraint Analysis

Constraints are prioritised so that the cheapest, most restrictive checks come first.

Unary constraints first (C1, C4). They shrink the domains before any search begins:

- D1: {M, A, N} minus N gives **{M, A}**
- D3: {M, A, N} minus M gives **{A, N}**
- D2: no unary constraint, so it stays **{M, A, N}**

Binary constraint (C2). Since D3 ∈ {A, N} and D2 must be earlier than D3, D2 cannot take Night (Night is the last shift, so nothing can come after it for D3). This is checked during search.

All-different constraint (C3). It is checked every time a value is assigned, to prevent two doctors sharing a shift.

The search space is therefore reduced from 3 × 3 × 3 = 27 combinations (or 6 permutations) to only a few candidates.

3. Backtracking Search

Method: Depth-first assignment of one variable at a time. After each assignment, all constraints whose variables are assigned are checked. If any constraint fails, the algorithm backtracks and tries the next value.

Variable order: D1, D2, D3. **Value order:** M, A, N.

Step-by-step trace

Step	Assignment tried	Constraint checks	Result
1	D1 = M	C1: M ≠ N ✓	Accept
2	D1 = M, D2 = M	C3: M already used by D1 ✗	Reject, try next value
3	D1 = M, D2 = A	C3: M, A distinct ✓; C2: D3 not yet assigned, no violation so far	Accept
4	D1 = M, D2 = A, D3 = M	C4: D3 ≠ M ✗; C3: M already used ✗	Reject, try next value
5	D1 = M, D2 = A, D3 = A	C3: A already used by D2 ✗	Reject, try next value
6	D1 = M, D2 = A, D3 = N	C1 ✓ (D1 = M); C2: A < N ✓; C3: M, A, N all distinct ✓; C4: N ≠ M ✓; C5: all assigned ✓	**Accept: solution found**

Full search tree (to show the other branches)

If the search continues after the first solution, the remaining branches are:

Branch	Outcome
D1 = M, D2 = N	C2 fails for any D3: D3 would have to be A (since M and N are used), but N < A is false. **Dead end, backtrack.**
D1 = A, D2 = M, D3 = N	C1 ✓, C2: M < N ✓, C3 ✓, C4 ✓. **Valid (second solution).**
D1 = A, D2 = N	D3 would have to be M, which violates C4, and N < M violates C2. **Dead end.**
D1 = N	C1 ✗. **Pruned immediately.**

4. Final Valid Schedule

The first solution found by backtracking is:

Shift	Doctor
Morning	**D1**
Afternoon	**D2**
Night	**D3**

Verification against all constraints:

- C1: D1 works Morning, not Night ✓
- C2: D2 (Afternoon) works before D3 (Night) ✓
- C3: one doctor per shift ✓
- C4: D3 works Night, not Morning ✓
- C5: each doctor has exactly one shift ✓

Note that a second valid schedule also exists: **D2 = Morning, D1 = Afternoon, D3 = Night**. Both satisfy every constraint. The first one is reported as the answer because backtracking returns the first consistent complete assignment under the chosen variable and value ordering.

5. Justification

- **Why backtracking:** it builds the solution incrementally and abandons a partial assignment as soon as a constraint is violated, so it avoids exploring invalid complete assignments.
- **Why check constraints early:** applying the unary constraints first and testing C3 at every assignment cuts the tree before it grows (for example, D1 = N was never explored, and D2 = N led to an immediate dead end).
- **Why the answer is correct:** every constraint was verified on the final assignment. The exhaustive tree also shows only two assignments are possible, and both have D3 on the Night shift, which is forced by C2 and C4 together.

6. Conclusion

Using backtracking search on the CSP, the valid hospital schedule is **Morning: D1, Afternoon: D2, Night: D3**, and all five constraints are satisfied.